

TRAVERS AGCM, AB, Canada

WMO: 718480

Lat: 50.3038N

Lon: 112.8626W

Elev: 955

StdP: 90.37

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.8	-24.7	-30.9	0.2	-27.5	-27.9	0.3	-24.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	30.3	15.4	28.3	14.9	26.5	14.8	18.0	24.8	16.9	24.4	16.0	23.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	12.3	21.0	14.3	11.4	19.9	13.1	10.6	18.5	54.6	24.5	51.3	24.4	48.3	23.9	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-31.5	33.8	3.6	2.6	-34.1	35.6	-36.2	37.1	-38.2	38.6	-40.8	40.5		
(5)			-31.7	20.1	3.5	1.2	-34.2	21.0	-36.3	21.7	-38.3	22.4	-40.9	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-6.0	-7.4	-1.1	4.7	10.8	15.1	18.0	17.6	12.8	5.8	-1.2	-7.6		
	DBStd	11.12	8.19	8.65	7.58	5.21	4.51	3.32	2.74	3.44	4.99	5.45	7.38	8.29		
	HDD10.0	2602	495	488	344	168	44	2	0	1	28	148	338	546		
	HDD18.3	4873	753	722	602	408	235	107	39	54	176	387	585	804		
	CDD10.0	852	0	0	1	10	70	154	248	237	111	19	3	0		
	CDD18.3	81	0	0	0	0	2	9	29	32	9	0	0	0		
	CDH23.3	1327	0	0	0	8	71	146	451	493	156	3	0	0		
	CDH26.7	374	0	0	0	1	12	33	124	160	43	1	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	350	12	11	17	30	53	71	43	39	29	17	16	12		
	PrecMax	534	26	24	57	68	92	201	102	94	110	47	38	28		
	PrecMin	159	1	1	1	8	13	23	7	1	2	3	1	1		
	PrecStd	67	6	6	11	15	22	38	26	22	23	11	10	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	11.7	16.7	24.5	28.3	30.5	32.6	33.2	30.4	22.9	18.2	10.4		
		MCWB	5.4	5.5	7.2	10.9	13.0	16.8	16.2	15.9	14.3	11.4	8.9	4.0		
	2%	DB	7.6	8.1	13.9	19.8	25.3	27.0	29.8	30.4	27.7	19.6	13.6	7.3		
		MCWB	3.1	3.1	6.0	8.6	12.1	14.6	16.2	15.3	13.5	9.9	6.6	2.9		
	5%	DB	5.8	5.5	11.4	16.6	22.4	24.7	27.6	28.3	24.8	17.0	10.7	5.2		
		MCWB	2.2	1.5	4.8	7.1	11.0	14.2	15.9	15.2	12.6	8.7	5.4	1.5		
	10%	DB	4.2	3.5	8.7	13.7	20.1	22.7	25.9	26.1	21.8	14.4	8.1	3.5		
		MCWB	1.3	0.5	3.4	5.6	10.0	13.0	15.7	14.9	11.6	7.3	3.5	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	6.0	7.9	10.9	13.6	18.2	19.8	18.4	15.7	12.4	9.2	5.4		
		MCDB	9.7	11.4	14.5	22.5	25.1	25.9	25.3	26.5	26.0	20.5	17.2	9.2		
	2%	WB	3.4	3.4	6.5	9.1	12.7	16.4	18.5	17.1	14.4	10.7	7.3	3.2		
		MCDB	7.1	7.7	13.0	18.2	23.4	24.0	25.2	26.0	24.5	18.0	12.9	6.8		
	5%	WB	2.5	1.8	5.1	7.5	11.7	15.1	17.6	16.3	13.2	9.1	5.5	1.8		
		MCDB	5.5	5.1	11.1	15.5	20.8	22.7	24.1	25.0	23.0	16.1	10.2	4.7		
	10%	WB	1.4	0.5	3.6	6.1	10.7	14.0	16.6	15.5	12.2	7.8	3.8	0.5		
		MCDB	4.2	3.4	8.3	13.1	18.3	20.5	23.5	23.7	20.8	13.9	7.8	3.5		
(35) Mean Daily Temperature Range	MDBR		9.2	10.1	10.8	12.2	12.4	12.0	13.7	14.0	13.0	11.9	10.0	9.1		
	5% DB	MCDBR	9.8	12.2	14.2	17.7	16.5	15.4	17.3	18.2	17.7	15.3	13.0	10.2		
		MCWBR	6.6	8.1	7.9	8.5	6.9	6.4	6.7	6.4	6.8	7.5	7.5	7.0		
	5% WB	MCDBR	9.3	11.7	13.7	16.9	14.6	13.9	14.4	15.5	16.4	14.7	12.9	10.8		
		MCWBR	6.6	8.2	7.9	8.4	6.3	6.5	6.8	6.3	6.7	7.6	7.8	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.260	0.282	0.322	0.338	0.334	0.336	0.351	0.312	0.279	0.254	0.234		
	taud		2.444	2.487	2.483	2.378	2.384	2.443	2.452	2.389	2.476	2.541	2.501	2.446		
	Ebn at Noon		818	904	936	923	918	922	914	881	888	868	811	792		
	Edh at Noon		58	72	87	109	113	108	105	107	87	68	54	49		
(44) All-Sky Solar Radiation	RadAvg		1.22	2.21	3.52	4.83	5.77	6.34	6.91	5.52	3.91	2.42	1.32	0.93		
	RadStd		0.10	0.20	0.27	0.33	0.42	0.41	0.34	0.35	0.35	0.27	0.12	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page